AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M238
Data Visualization

Final Examination
Spring Semester 2025/2026

Date: 11 / .June /2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **TWO** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1

TRUE or FALSE **(15 marks)**

MARK the following statements are TRUE or FALSE

1. **A timeline is useful for showing events over time and can support storytelling by highlighting important dates and patterns.**
2. **Bar-in-bar charts are useful for comparing a measure against a goal or comparing two measures starting from the same zero baseline.**
3. **Timelines can be created directly in Tableau as a built-in chart type without any preparation.**
4. **Stacked bar charts are always the best option for precise quantitative comparison because all bars start from zero.**
5. **Likert scales are commonly used in surveys to measure sentiment and tendencies.**
6. **Likert-scale data can only be visualized using one chart type in Tableau.**
7. **The $PROPER()$ function in Tableau helps format text fields by converting them to proper case.**
8. **Sequential color is best used when data contains both positive and negative values around a midpoint.**
9. **When used properly, color can enhance and clarify a data visualization.**
10. **Tableau uses the same type of color palette for both discrete and continuous fields.**
11. **Sequential color encodes quantitative values from low to high using gradients of a single color.**
12. **Diverging color uses a meaningful midpoint (such as zero) to separate positive and negative values.**
13. **Stepped colors group continuous values into a fixed number of color-coded bins.**
14. **Reversing colors in a sequential palette makes higher values darker instead of lower values.**
15. **Categorical colors are used to encode fields that have no inherent quantitative order.**

PART 2

ALL QUESTIONS ARE REQUIRED **(35 marks)**

Question 1 (7 Marks)

A. **How can you prepare your date field in Tableau before creating a timeline visualization? (1 Marks)**
B. **What are the two most common visualization types for Likert-scale data? (4 Marks)**

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C. What does the $PROPER()$ function do to a text string in Tableau?. (2 Marks)

Question 2 (7 Marks)

A. What are the three primary ways to encode data with color in data visualization? (3 Marks)
B. When should sequential color be applied in data visualization? (2 Marks)
C. How are positive and negative profit values represented using diverging color in the example? (2 Marks)

Question 3 (6 Marks)

A. How do stepped colors and reversed colors change the way data is encoded with color? (2 Marks)
B. What types of data are best represented using categorical colors? (2 Marks)
C. Why should you test a visualization with and without mark borders? (2 Marks)

Question 4 (8 Marks)

A. What are the two most common visualization types for Likert-scale data? (2 Marks)
B. How can you improve a basic 100% stacked bar chart for visualizing Likert-scale data? (4 Marks)
C. What two chart types are combined to create a lollipop chart? (2 Marks)

Question 5 (7 Marks)

A. What is the main purpose of using a word cloud in a presentation or visualization? (2 Marks)
B. What are the five steps to visual data storytelling? (5 Marks)

End of questions

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